

AD 502 – Machine Learning

Pre-Reading Material – Lecture 11

Bias Variance Tradeoff

Estimated Reading Time: 25–30 minutes

Why This Topic Matters

A machine learning model should perform well not only on the training data but also on **unseen data**. The bias–variance trade-off helps us understand why a model may be too simple, too complex, or appropriately balanced.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain **bias** and **variance** in machine learning.
- Distinguish between **underfitting** and **overfitting**.
- Explain the bias–variance trade-off.
- Understand the relationship between model complexity and prediction error.
- Identify techniques used to reduce high bias or high variance.

Activate Your Prior Knowledge

1. Very Simple Model

Suppose a straight-line model is used to represent data that clearly follows a curved pattern.

Think: Can a model that is too simple capture the underlying relationship?



2. Very Complex Model

Suppose a model fits every training point almost perfectly but performs poorly on new data.

Think: Is the model learning the actual pattern or memorizing the training data?

3. Model Selection

Two models have similar training performance, but one performs much better on unseen test data.

Think: Which model should we prefer and why?

Prerequisite Concepts

You should be familiar with:

- Supervised learning
- Training and testing datasets
- Regression and classification
- Model complexity
- Mean Squared Error (MSE)
- Overfitting and underfitting

Core Concepts

1. Bias

Bias is the error introduced when a model makes overly strong assumptions about the underlying relationship in the data. A high-bias model is generally too simple and may fail to capture important patterns.



2. Variance

Variance represents how much a model's predictions change when it is trained on different samples of the training data. A high-variance model is highly sensitive to the particular training dataset.

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

3. Low Bias and High Bias

A model with **low bias** can represent complex relationships effectively. A model with **high bias** is usually too restrictive and tends to make systematic errors.

4. Low Variance and High Variance

A model with **low variance** produces relatively stable predictions across different training datasets. A model with **high variance** can change substantially when the training data changes slightly.

5. Underfitting

Underfitting occurs when a model is too simple to capture the important patterns in the training data. It generally results in **high training error and high testing error**, corresponding to high bias.

6. Overfitting

Overfitting occurs when a model learns the training data too closely, including noise and random fluctuations. It generally produces **very low training error but high testing error**, corresponding to high variance.



7. Bias–Variance Trade-off

The bias–variance trade-off describes the balance between two sources of prediction error:

- **High Bias:** Model is too simple → underfitting.
- **High Variance:** Model is too complex → overfitting.
- **Balanced Model:** Appropriate complexity → better generalization.

8. Model Complexity

As model complexity increases, bias generally decreases while variance tends to increase. The objective is to select an appropriate level of complexity that provides good performance on unseen data.

9. Generalization

Generalization is the ability of a machine learning model to perform well on new, unseen data. A good model learns the underlying patterns rather than simply memorizing the training examples.

10. Reducing High Bias

High bias can often be reduced by:

- Using a more complex model
- Adding useful features
- Reducing excessive regularization
- Improving the representation of the data

11. Reducing High Variance

High variance can often be reduced by:

- Collecting more training data
- Using a simpler model
- Applying regularization
- Feature selection
- Using ensemble methods such as bagging

12. Cross-Validation

Cross-validation evaluates a model using multiple training and validation splits. It provides a more reliable estimate of how well a model is likely to generalize to unseen data and can help in selecting an appropriate model complexity.

Guided Reading Questions

1. What is bias in machine learning?
2. What is variance?
3. What is underfitting?
4. What is overfitting?
5. How are bias and variance related to model complexity?
6. What is the bias–variance trade-off?
7. Why is generalization important?
8. How can high bias be reduced?
9. How can high variance be reduced?
10. How does cross-validation help in model selection?

Reflection Activity

Write 100–150 words explaining why a model with **very low training error** may not necessarily be the best machine learning model. Discuss overfitting, variance, and performance on unseen data.



Self-Assessment

1. A model that is too simple generally suffers from high _____.
2. A model that is highly sensitive to training data generally has high _____.
3. High bias is commonly associated with _____.
4. High variance is commonly associated with _____.
5. A model that memorizes training examples is likely to suffer from _____.
6. The ability to perform well on unseen data is called _____.
7. Increasing model complexity generally decreases bias and increases _____.
8. Regularization is commonly used to reduce _____.
9. Collecting more training data can help reduce _____.
10. Cross-validation helps estimate a model's ability to _____ to unseen data.

Key Takeaway

The **bias-variance trade-off** is about finding the right balance between a model that is too simple and one that is too complex. A well-designed machine learning model should capture meaningful patterns while avoiding both **underfitting and overfitting**, ultimately achieving good **generalization** on unseen data.

